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COURSE CODE: SBT-DF202
LAB NAME: STEGANOGRAPHY AND HIDDEN DATA DETECTION
DATE: 05/09/2026
INSTRUCTOR: AMINU IDRIS

Laboratory instructions

Before beginning this lab, review the **Module 8 lessons and study material on Steganography and Hidden Data Analysis**. You must understand the concepts of hidden information, substitution, least significant bits, bitmap images, Steghide, extraction and password recovery before performing the practical tasks.

Required Carrier Image

Download the authorised lab BMP image:

Steganography Lab Carrier Image

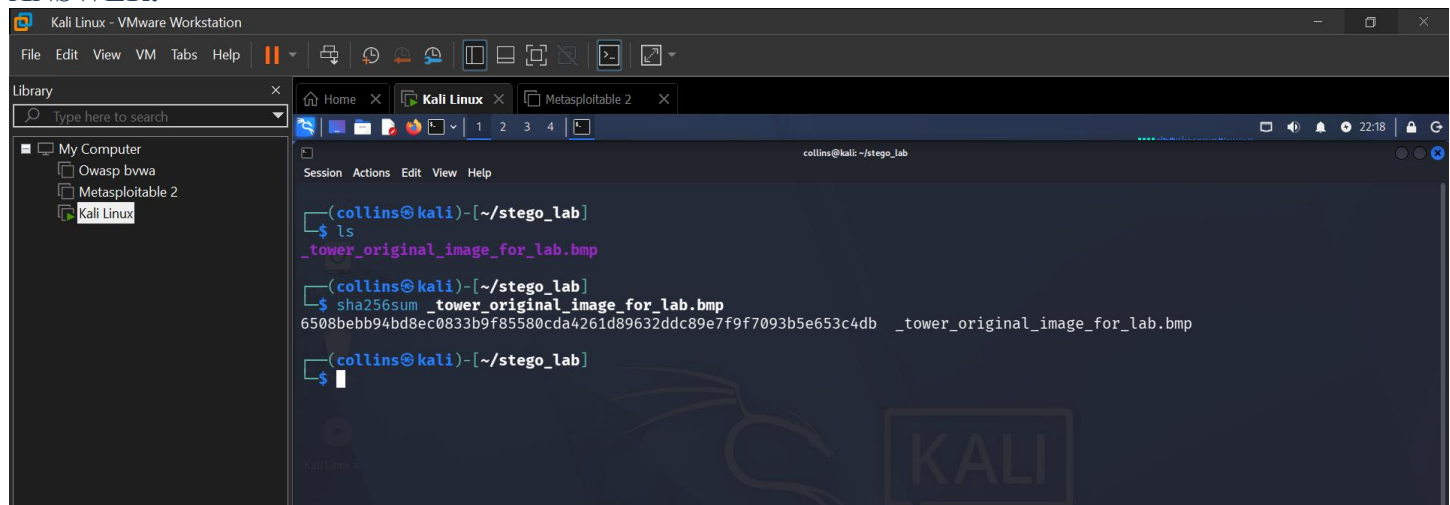
If browser download is required, use the link above. The source practical identifies this specific image as the lab carrier.

Part A — Prepare and Preserve the Evidence

Create a dedicated working folder, download the carrier image, confirm its file type and calculate its SHA-256 hash.

Preserve the original downloaded image. Do not overwrite it during the embedding exercise.

ANSWER:



```
(collins@kali) - [~/stego_lab]
$ ls
_tower_original_image_for_lab.bmp

(collins@kali) - [~/stego_lab]
$ sha256sum _tower_original_image_for_lab.bmp
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db _tower_original_image_for_lab.bmp

(collins@kali) - [~/stego_lab]
$
```

Part B — Create the Secret Evidence File

Create a text file named:

secret.txt

Include:

- Your full name
- A short controlled evidence message
- Confirmation that the file was created for the ICDFA lab

Calculate and record the SHA-256 hash of the file.

ANSWER:

```

Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bvwa
Metasploitable 2
Kali Linux

(collins@kali) - [~/stego_lab]
$ ls
secret.txt  _tower_original_image_for_lab.bmp

(collins@kali) - [~/stego_lab]
$ cat secret.txt
COLLINS AMADI
This is a controlled evidence message for the steganography lab.
This file was created for the ICDFA lab.

(collins@kali) - [~/stego_lab]
$ sha256sum secret.txt
417df3ef107cdb8d59db9efbb3a77016ddba1a97a67cdb955dd1c04eb43f6586  secret.txt

(collins@kali) - [~/stego_lab]
$

```

Part C — Embed the Hidden File with Steghide

Use Steghide to embed secret.txt into a new stego image.

For the guided portion of the lab, use the lab password:

1234

The source lab uses this password specifically so the later dictionary-recovery task can be completed.

Example structure:

```
steghide embed \ -ef secret.txt \ -cf tower_original_image_for_lab.bmp \ -sf tower_stego_lab4.bmp \ -p 1234
```

Calculate and record the hash of the resulting stego image.

ANSWER:

```

Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bvwa
Metasploitable 2
Kali Linux

(collins@kali) - [~/stego_lab]
$ steghide embed -ef secret.txt -cf tower_original_image_for_lab.bmp -sf tower_stego_lab4.bmp -p 0436 -f
embedding "secret.txt" in "tower_original_image_for_lab.bmp"... done
writing stego file "tower_stego_lab4.bmp"... done

(collins@kali) - [~/stego_lab]
$

```

Part D — Extract and Verify the Hidden Evidence

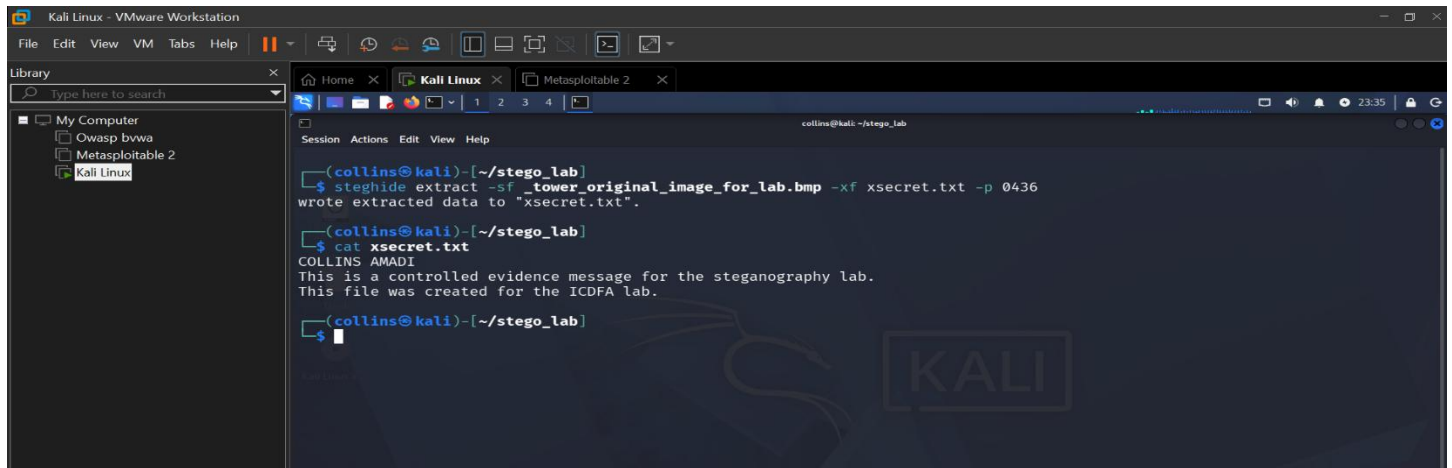
Extract the concealed file using the correct password.

Example:

```
mkdir -p extracted steghide extract \-sf tower_stego_lab4.bmp \-xf extracted/extracted_secret.txt \-p 1234
```

Compare the original and extracted files using diff and/or SHA-256 hashes. The recovered file should match the original if extraction is successful.

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwva
  Metasploitable 2
  Kali Linux

(collins@kali)-[~/stego_lab]
$ steghide extract -sf _tower_original_image_for_lab.bmp -xf xsecret.txt -p 0436
wrote extracted data to "xsecret.txt".

(collins@kali)-[~/stego_lab]
$ cat xsecret.txt
COLLINS AMADI
This is a controlled evidence message for the steganography lab.
This file was created for the ICDA lab.

(collins@kali)-[~/stego_lab]
$
```

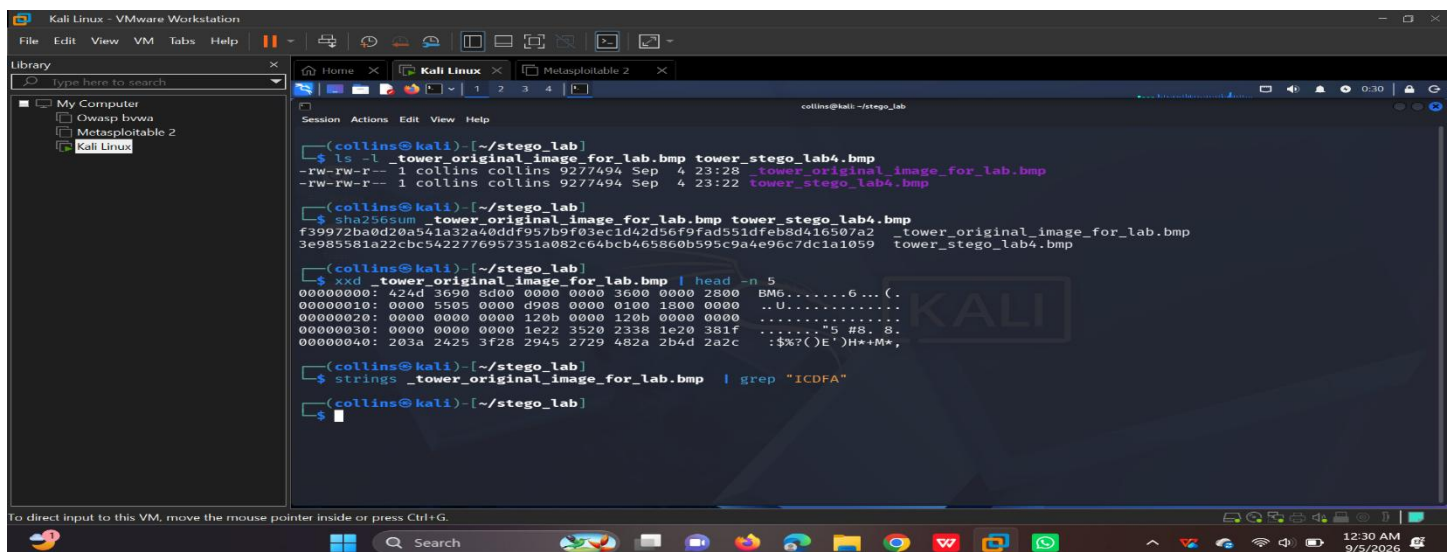
Part E — Compare Original and Stego Images

Compare:

- File sizes
- SHA-256 hashes
- Initial bytes using xxd
- Readable strings where applicable

Explain why two images may look visually similar while their underlying digital content and hashes are different.

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwva
  Metasploitable 2
  Kali Linux

(collins@kali)-[~/stego_lab]
$ ls -l _tower_original_image_for_lab.bmp tower_stego_lab4.bmp
-rw-rw-r-- 1 collins collins 9277494 Sep  4 23:28 _tower_original_image_for_lab.bmp
-rw-rw-r-- 1 collins collins 9277494 Sep  4 23:22 tower_stego_lab4.bmp

(collins@kali)-[~/stego_lab]
$ sha256sum _tower_original_image_for_lab.bmp tower_stego_lab4.bmp
f39972ba9d20a541a32a40ddf957b9f03ec1d42d56f9fad551df8b8d416507a2  _tower_original_image_for_lab.bmp
3e985581a22c5422776957351a082c64bcb465860b595c9a4e96c7d1a1059  tower_stego_lab4.bmp

(collins@kali)-[~/stego_lab]
$ xxd _tower_original_image_for_lab.bmp | head -n 5
00000000: 424d 3690 8d00 0000 0000 3600 0000 2800  BM6.....6... (.
00000010: 0000 5505 0000 d908 0000 0100 1800 0000  ..U.....
00000020: 0000 0000 0000 120b 0000 120b 0000 0000  .....
00000030: 0000 0000 0000 1e22 3520 2338 1e20 381f  .....5 #8; 8
00000040: 203a 2425 3f28 2945 2729 482a 2b4d 2a2c  :;%?()E')H*+M*,

(collins@kali)-[~/stego_lab]
$ strings _tower_original_image_for_lab.bmp | grep "ICDA"
(collins@kali)-[~/stego_lab]
$
```

Part F — Controlled Password Dictionary Test

Create a small lab wordlist that contains the known password 1234.

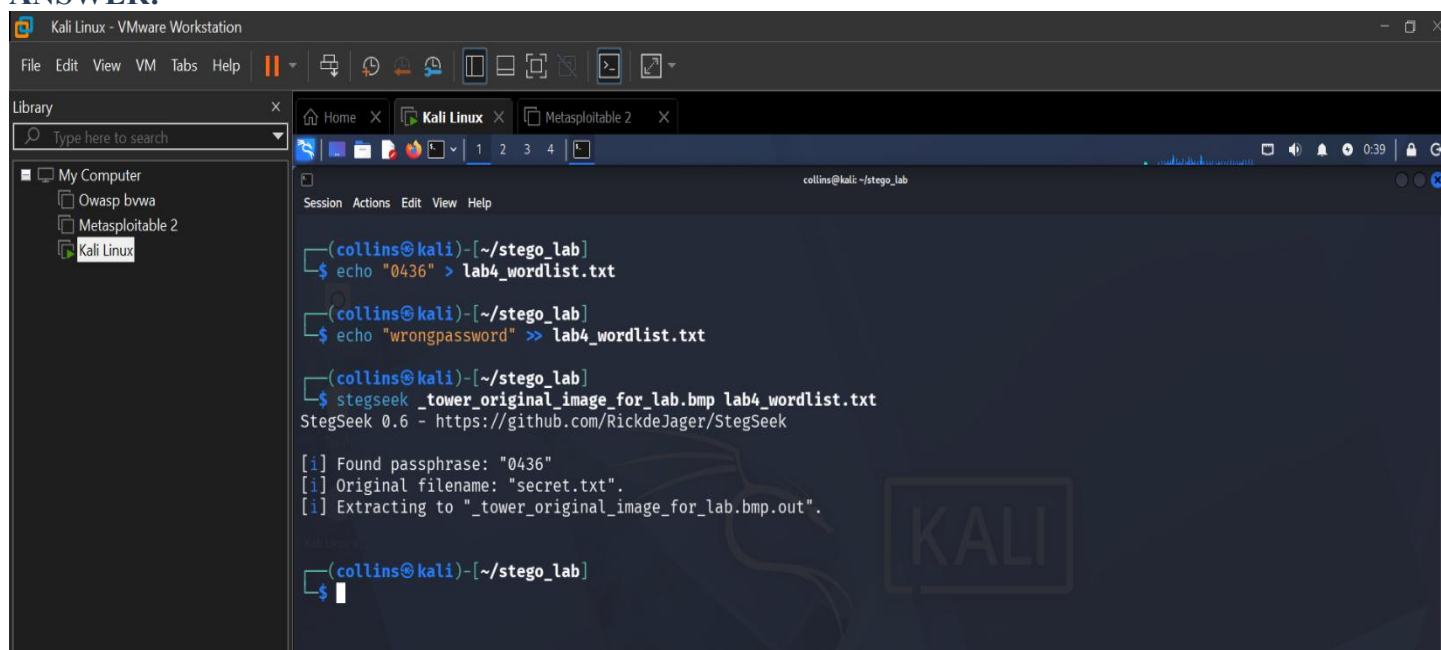
Use StegSeek against the stego image:

```
stegseek tower_stego_lab4.bmp lab4_wordlist.txt
```

Document the result.

If StegSeek is unavailable or fails in your environment, document the error clearly and complete the Steghide embedding/extraction workflow successfully instead.

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
  Owasp bwba
  Metasploitable 2
  Kali Linux

(collins@kali) - [~/stego_lab]
$ echo "0436" > lab4_wordlist.txt

(collins@kali) - [~/stego_lab]
$ echo "wrongpassword" >> lab4_wordlist.txt

(collins@kali) - [~/stego_lab]
$ stegseek _tower_original_image_for_lab.bmp lab4_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "0436"
[i] Original filename: "secret.txt".
[i] Extracting to "_tower_original_image_for_lab.bmp.out".

(collins@kali) - [~/stego_lab]
$
```

Part G — Independent Student Task

Create your own hidden evidence file:

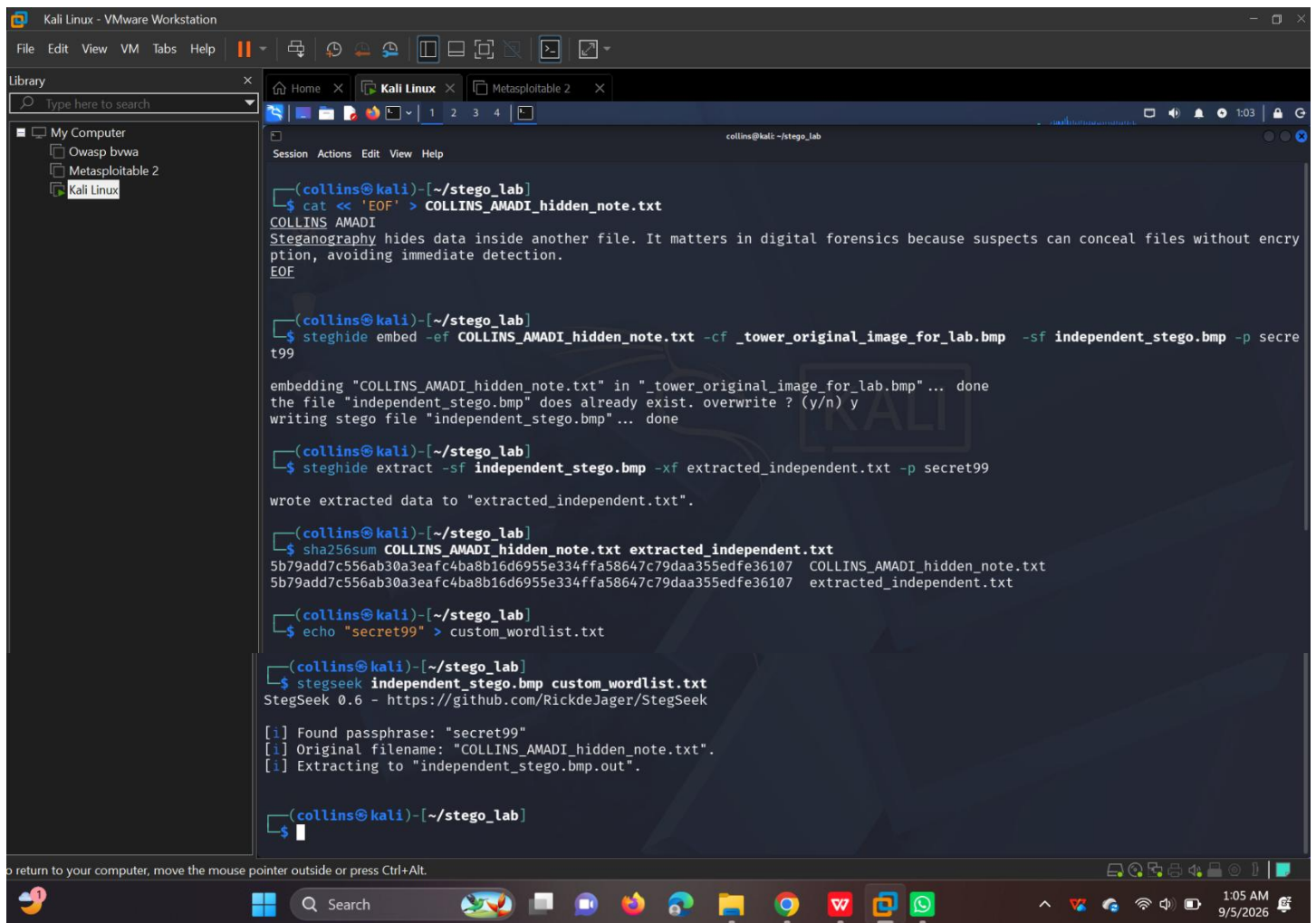
```
firstname_lastname_hidden_note.txt
```

Write a short explanation of what steganography is and why it matters in digital forensics.

Then:

1. Embed it into the carrier image.
2. Use a password of your choice.
3. Extract it again.
4. Verify integrity using hashes.
5. Create a small wordlist containing your chosen password.
6. Test the recovery workflow.
7. Document every step with screenshots.

ANSWER:



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the following commands and output:

```
(collins@kali)-[~/stego_lab]
$ cat << 'EOF' > COLLINS_AMADI_hidden_note.txt
COLLINS AMADI
Steganography hides data inside another file. It matters in digital forensics because suspects can conceal files without encryption, avoiding immediate detection.
EOF

(collins@kali)-[~/stego_lab]
$ steghide embed -ef COLLINS_AMADI_hidden_note.txt -cf _tower_original_image_for_lab.bmp -sf independent_stego.bmp -p secret99

embedding "COLLINS_AMADI_hidden_note.txt" in "_tower_original_image_for_lab.bmp"... done
the file "independent_stego.bmp" does already exist. overwrite? (y/n) y
writing stego file "independent_stego.bmp"... done

(collins@kali)-[~/stego_lab]
$ steghide extract -sf independent_stego.bmp -xf extracted_independent.txt -p secret99

wrote extracted data to "extracted_independent.txt".

(collins@kali)-[~/stego_lab]
$ sha256sum COLLINS_AMADI_hidden_note.txt extracted_independent.txt
5b79add7c556ab30a3eafc4ba8b16d6955e334ffa58647c79daa355edfe36107 COLLINS_AMADI_hidden_note.txt
5b79add7c556ab30a3eafc4ba8b16d6955e334ffa58647c79daa355edfe36107 extracted_independent.txt

(collins@kali)-[~/stego_lab]
$ echo "secret99" > custom_wordlist.txt

(collins@kali)-[~/stego_lab]
$ stegseek independent_stego.bmp custom_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "secret99"
[i] Original filename: "COLLINS_AMADI_hidden_note.txt".
[i] Extracting to "independent_stego.bmp.out".

(collins@kali)-[~/stego_lab]
$
```

The bottom of the screen shows the VMware Workstation interface with the taskbar and system tray. The taskbar includes icons for the Start menu, Search, and various applications. The system tray shows the time as 1:05 AM on 9/5/2026.